

Virtualization – Detailed Notes

What is Virtualization?

Virtualization is the process of creating a **virtual version** of something — such as a computer, operating system, storage device, or network — instead of using physical hardware.

It allows one physical machine (the **host**) to run multiple **virtual machines (VMs)**, each with its own operating system and applications.

Example

Suppose you have one computer with Windows 11.

By installing **VMware Workstation** or **Oracle VirtualBox**, you can create:

- One VM running **Windows 10**
- Another VM running **Kali Linux**

Both run **independently** on the same hardware — this is virtualization.

How Virtualization Works

1. **Physical Machine (Host):**
Your real computer hardware — CPU, memory, disk, etc.
2. **Hypervisor:**
Software that creates and manages virtual machines.
It allocates system resources to each VM.

Examples: VMware, Hyper-V, VirtualBox, KVM

3. Virtual Machines (Guest):

Each VM behaves like a full computer with:

- a. Virtual CPU, RAM, and disk
- b. Its own OS (like Kali Linux, Windows, Ubuntu)

The hypervisor ensures that each VM runs independently without interfering with others.

Types of Virtualization

Type	Description	Example
1. Hardware Virtualization	Creating virtual machines on a host system via a hypervisor.	VMware, Hyper-V
2. OS Virtualization (Containers)	Running isolated user spaces on a single OS kernel.	Docker, LXC
3. Server Virtualization	Splitting one physical server into multiple virtual servers.	VMware vSphere
4. Storage Virtualization	Pooling multiple storage devices into one virtual storage unit.	IBM SAN Volume Controller
5. Network Virtualization	Creating virtual networks independent of physical hardware.	VMware NSX
6. Desktop Virtualization	Centralized desktops accessible remotely.	Citrix, VDI

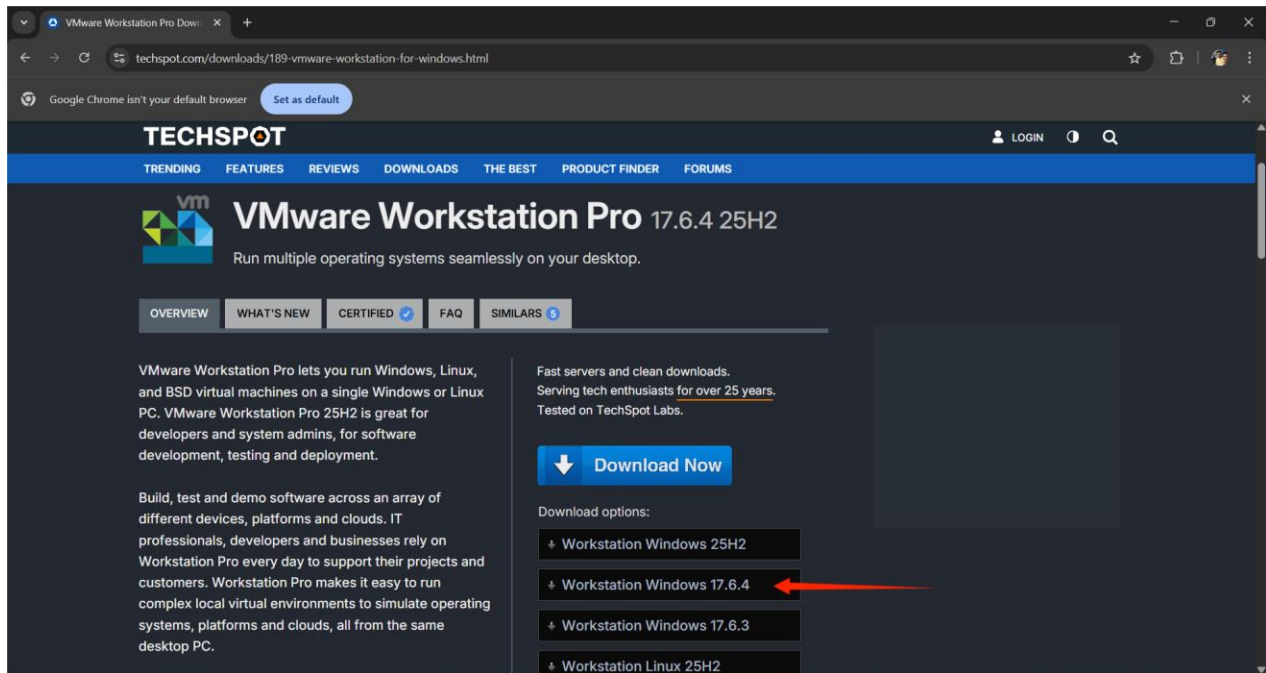
Advantages

- Better resource utilization
- Cost-effective (less hardware needed)
- Easier testing and development
- Isolation (safe from malware in host)
- Scalable and portable

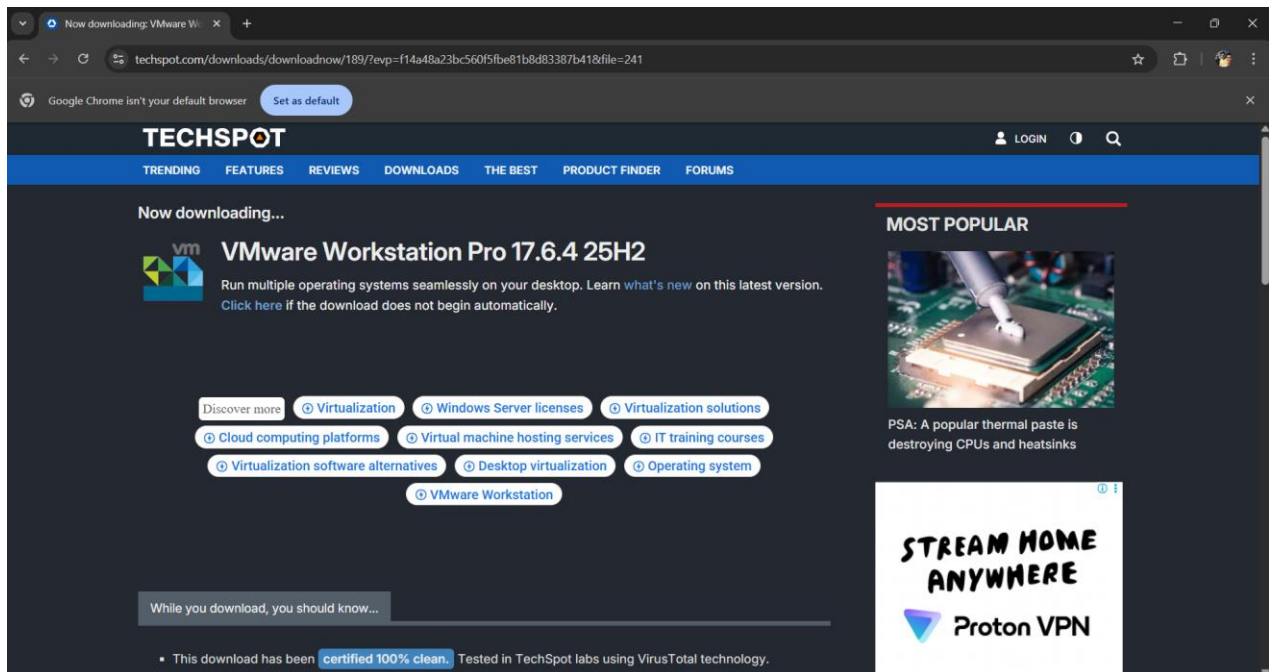
Installing VMware and Kali Linux

Step 1: Open your browser and go to the VMware Workstation website click on Techspot website show in below diagram.

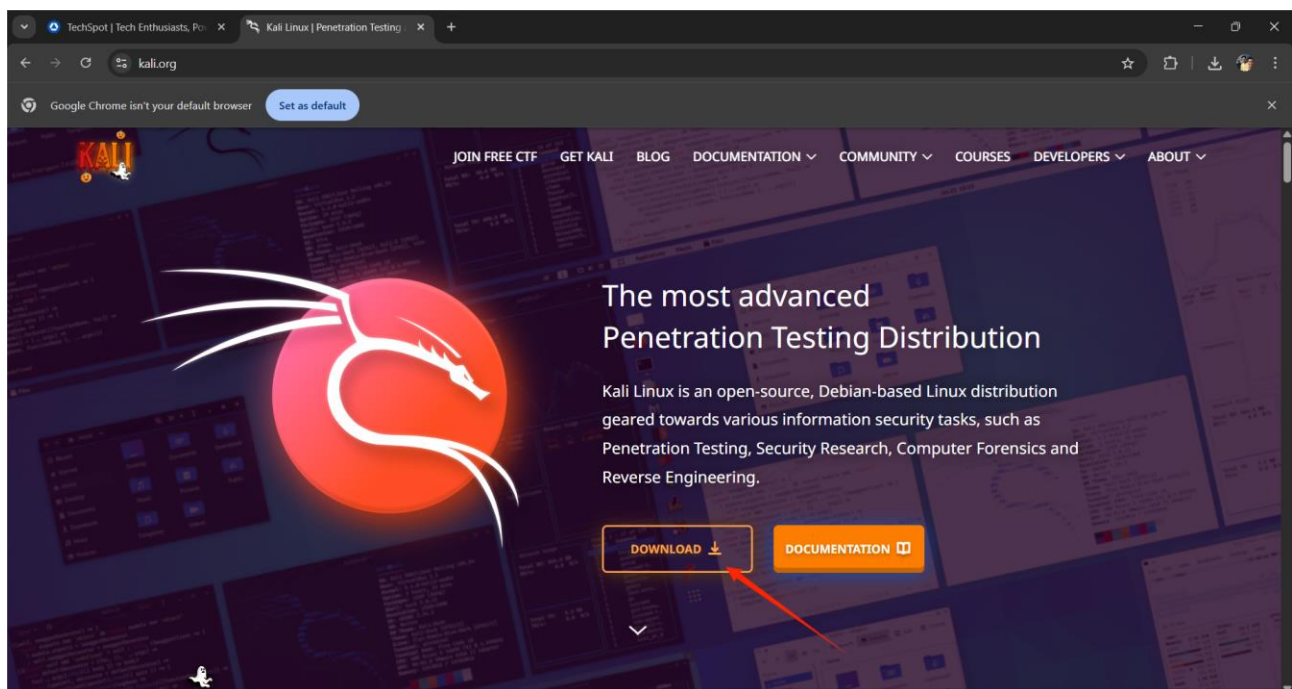
Step 2: Click on latest version of VMware Workstation after click on this show in diagram.



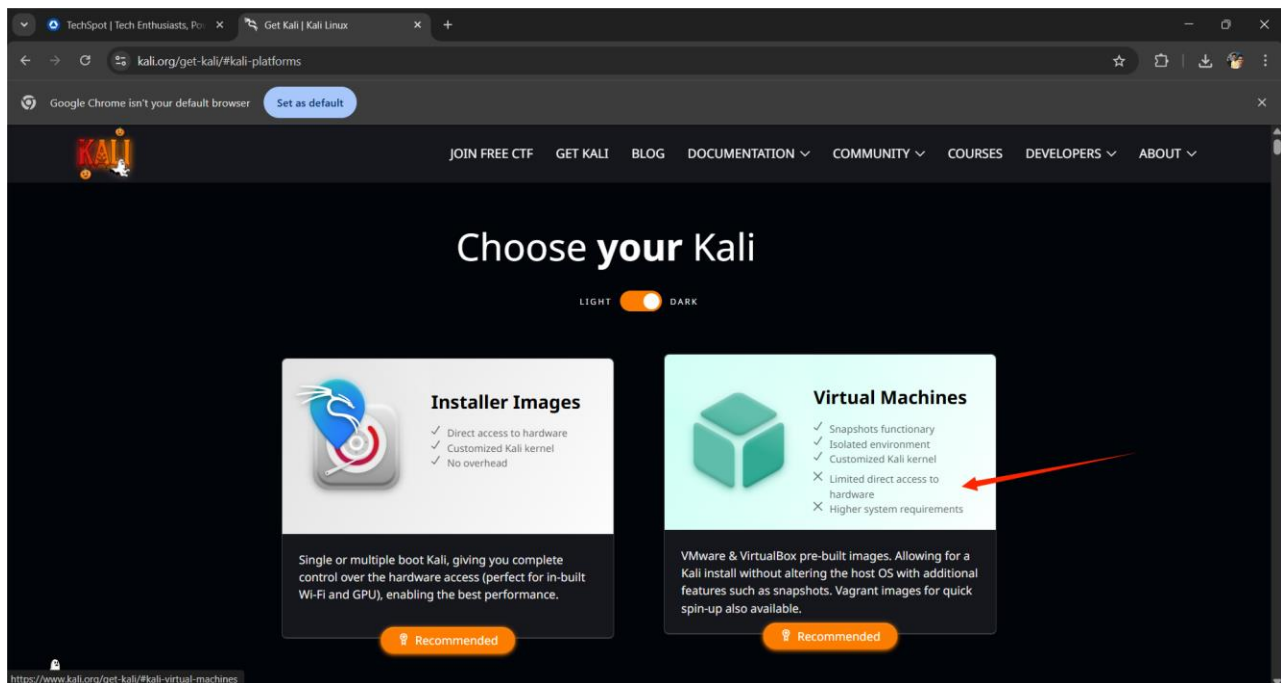
Step 3 : After click on this you see the new web page in this page you see the Downloading is start (show like this page).



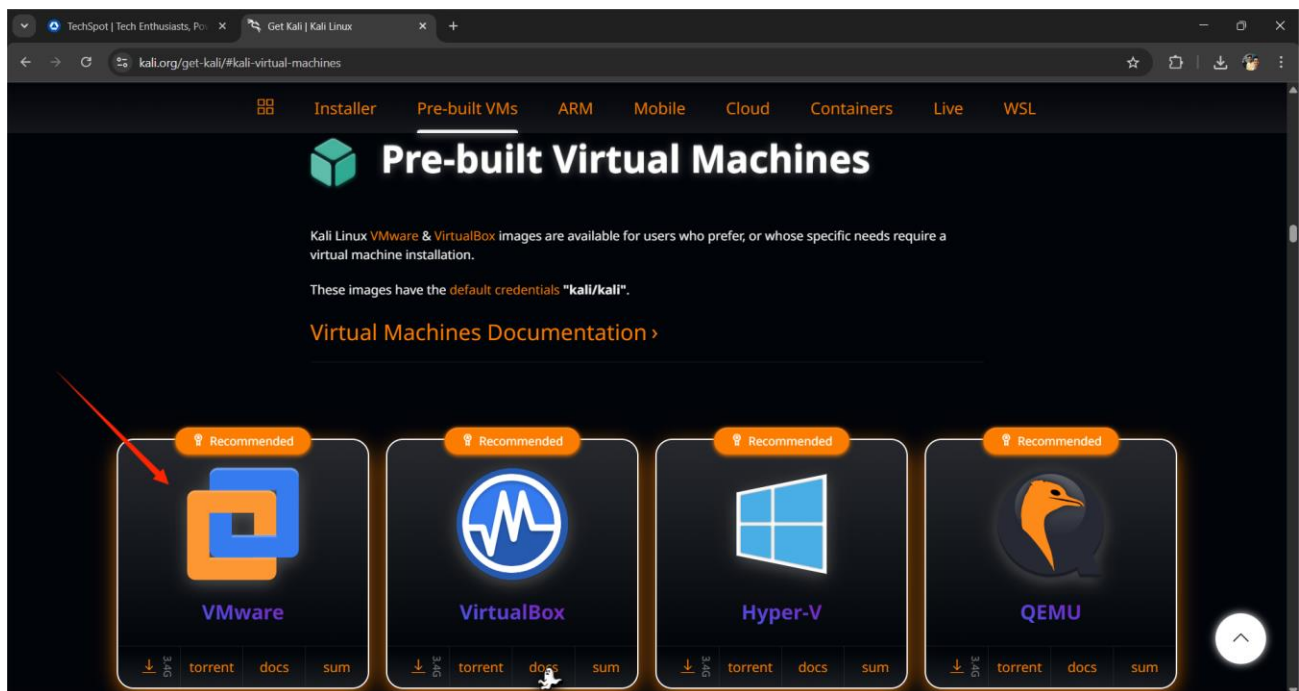
Step 4 : After this we download the Kali Linux on Kali.org site. This is original site of Kali Linux.



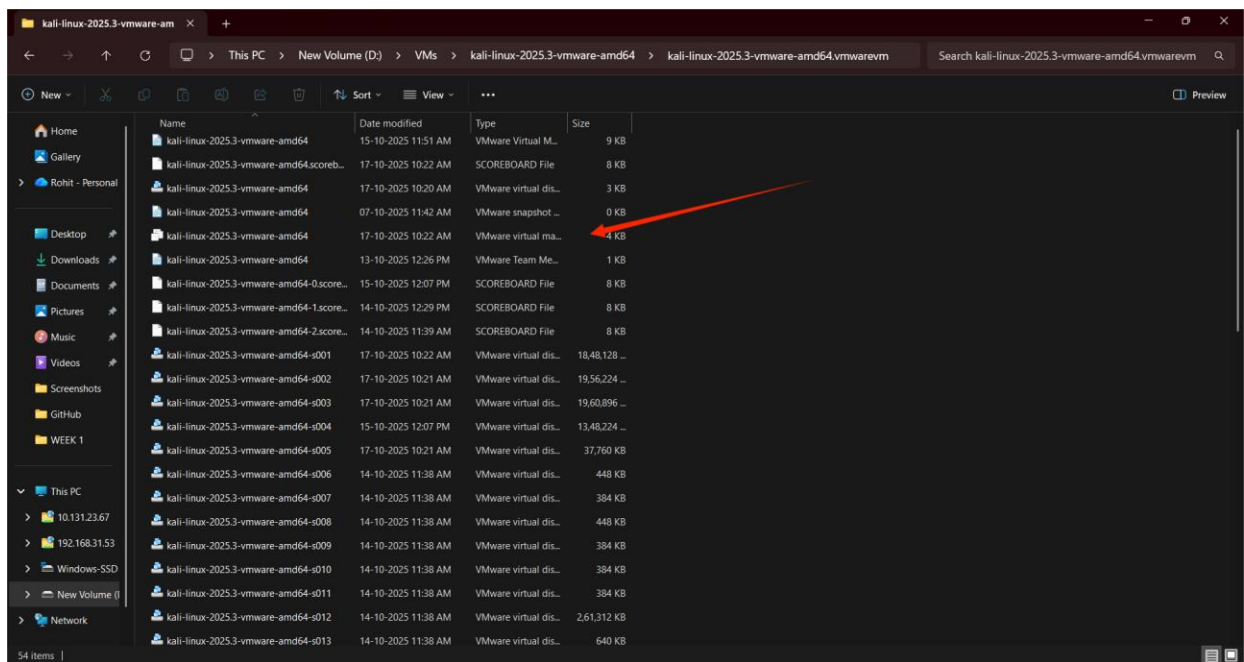
Step 5 : After open the Kali Linux site click on the download button. After that you see the this page click on Virtual Machines.



Step 6 : After this click on VMware to download this.



Step 7 : After complete both downloads file double click on VMware file to open this and setup this .After that extract the Kali Linux zip file and open file and click on .vmx extension show in below diagram.



Step 8 : After this you see like show in below diagram in this you setup Memory , Processors, You also change Network Adapter NAT into Bridges.

